

Tutorial 3

1. For each of the three DFAs at the end of the question sheet, find a *minimal* DFA which accepts the same language.
2. From the course textbook by Salling: Test 2.7, Exercises: 2.2 c, 2.3 b, 2.5, 2.7.
3. Consider the following language over $\Sigma = \{a, b\}$

$$L = \{w \in \{a, b\}^* : w \text{ contains at least three times as many } b\text{'s as } a\text{'s}\}.$$

- (a) Show that L is not regular.
 - (b) Is $\bar{L}$ regular?
4. Which of the following languages are regular, and which are not?

$$L_1 = \{uvu^{\text{rev}} : u, v \in \{a, b\}^* \text{ and } |u| = 2\}$$

$$L_2 = \{uvu^{\text{rev}} : u, v \in \{a, b\}^* \text{ and } |v| = 2\}$$

$$L_3 = \{w \in \{a, b\}^* : w \text{ has no occurrences of the substring } aabb\}$$

$$L_4 = \{w \in \{a, b\}^* : w \text{ has the same number of occurrences of the substring } aa \text{ as the substring } bb\}$$

5. From the course textbook by Salling: Test 2.8, Exercises 2.8 – 2.14.
6. In parts (a)-(c) below decide whether the language which is described is regular or not. Note \$1 is worth 100c (one-hundred cents).
 - (a) Suppose we have a change machine in which one may deposit 25c coins and 50c coins. and if the total is a whole number of dollars, you can push a button on the machine, and the machine spits out that number of dollar coins. If you push the button and there is not a whole number of dollars, all coins that you put in the machine are returned to you. Let the 50c coins be coded by 'a' and 25c coins coded by 'b'. Describe (in words) the language (over alphabet = $\{a, b\}$) of the coin sequences (input to the machine) which will result in the machine spitting out dollar coins. Is this language regular?
 - (b) Now suppose that the change machine can only give change for amounts of no more than \$100, so the machine accepts coins (a and b) whose total amount is a whole number of dollars and the total is at most \$100. Furthermore one needs to check that there is at least \$100 in the machine. One dollar (\$1) coins are coded by 'c'. The strings to be accepted are (exactly) those of the form xy such that $x \in \{a, b\}^*$, $y \in \{c\}^*$, the number of cents that x corresponds to is $\leq 10,000$ and evenly divisible by 100, and $|y| \geq 100$. Is this language regular?¹
 - (c) Lastly, suppose that the exchange machine can change any amount, provided that there are enough one dollar (\$1) coins (and that the a and b coins which have been deposited correspond to a whole amount of dollars.) The strings to be accepted are then of the form xy where $x \in \{a, b\}^*$, $y \in \{c\}^*$, the number of cents that x corresponds to is evenly divisible by 100 and the total amount in dollars is at most $|y|$ (= the number of one dollar (\$1) coins in the machine). Is this language regular?

¹Note: the y part corresponds to the supply of one dollar (\$1) coins in the change machine.

7. Suppose that languages L_1 and L_2 are regular, and that language L_3 is not regular. Which of the following hold (for all such choices of language)?

- (a) L_1L_2 is regular.
- (b) L_1L_3 is regular.
- (c) L_1L_3 is not regular.
- (d) $\overline{L_1}$ is regular.
- (e) $\overline{L_3}$ is regular.
- (f) $\overline{L_3}$ is not regular.
- (g) $L_1 \cup L_2$ is regular
- (h) $L_1 \cap L_3$ is regular.
- (i) $L_1 \cup L_3$ is not regular.
- (j) $L_1 \setminus L_2$ is regular.²
- (k) $L_1 \setminus L_3$ is regular.

²Hint: $L_1 \setminus L_2 = L_1 \cap \overline{L_2}$.